# SAI BONTHA

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## EXPERIENCE

**Embedded Software Engineer** Jan 2026 – Present

*Dell Technologies* *Austin, TX*

- Design and optimize **multithreaded Embedded Linux application infrastructure** for **high-throughput data paths**, applying task scheduling, synchronization, and rate-limiting policies to sustain **SLOs** under load
- Perform low-level debugging with **GDB** and **JTAG/SWD**, including **PCIe/NVMe data-path validation**, kernel and driver-level analysis, and root-cause of runtime faults
- Develop and enhance **Python automation** for test execution, log processing, and validation, plus Bash for build and bring-up workflows — improving selected workflows by **10%** and reducing recurring debugging effort by **8%**
- Apply **fault-tolerant design** across redundant paths, including **failover**, **replication**, and self-healing retry logic, to preserve correctness during network faults and brownouts
- Establish **KPIs** and build proactive **telemetry dashboards and alerts** to surface production issues and reduce mean-time-to-detection
- Support integration and troubleshooting of **SPI**, **I2C**, **UART**, and **TCP/IP** paths and hardware/software system bring-up with remote lab equipment

**Embedded Software Engineer** Jun 2020 – Nov 2023

*HCLTech* *Hyderabad, India*

- Developed and maintained **embedded software modules** in C/C++ for firmware and real-time software environments, including **RTOS-based applications** with task scheduling and synchronization
- Diagnosed communication and integration issues involving **CAN**, **SPI**, **I2C**, and **UART** by analyzing messages, interface behavior, logs, and execution flow
- Used **GDB**, **JTAG**, and **SWD** plus Linux kernel and device-driver analysis to investigate firmware issues, runtime faults, and memory-related problems
- Developed **Python and Perl scripts** to automate testing, log analysis, validation, and regression activities in a **CI/CD** environment
- Supported **MQTT-based communication** workflows (publish/subscribe, payload handling, reconnect scenarios) with load-shedding policies for resilience; improved selected RTOS execution paths by **13%** and automated test coverage to **85%**

## PROJECTS

**Embedded Linux IoT Gateway & Telemetry System | C/C++, MQTT**

- Built a **multithreaded Embedded Linux gateway** collecting device data and transmitting telemetry over **MQTT**, with synchronization and thread-safe queues for high-throughput ingestion
- Implemented C/C++ modules for device communication and data processing, including low-level **PCIe/NVMe data transfers** to storage, and integrated **CAN**, **SPI**, **I2C**, and **UART** data collection
- Developed **Python scripts** for automated test execution, payload validation, log parsing, and regression testing, with unit and integration coverage of communication and error-handling scenarios

## OPEN SOURCE

**Zephyr RTOS | C, I2S**

- Upstream PR: **config_get** for the **LiteX/MAX32 I2S drivers** (approved by maintainers, merged); issue root-causing the **ESP32 DVP driver** frame-production stall, confirmed on ESP32-S3 hardware

**scpi-parser | C, Test Instrumentation**

- Upstream pull request: **SCPI_NO_HIDDEN_VISIBILITY** opt-out fixing **-Werror** build breaks on bare-metal toolchains

## EDUCATION

**Youngstown State University** Youngstown, OH

*M.S. Electrical and Computer Engineering* *Jan 2024 – Dec 2025*

## TECHNICAL SKILLS

**Languages:** C, C++, Python, Perl, Bash, Shell Scripting
**Embedded:** Embedded Linux, RTOS, Firmware Development, Real-Time Systems, Linux Kernel/Driver Programming, Multithreading, Virtualization (SRIOV, VirtIO), Device Drivers, BSP, Board Bring-Up, U-Boot, Device Tree
**Protocols:** CAN, SPI, I2C, UART, MQTT, TCP/IP, PCIe, NVMe, DPDK
**Debugging & Testing:** GDB, JTAG, SWD, Log Analysis, Root-Cause Analysis, Fault Isolation, Fault Injection, Kernel Oops Analysis, Unit/Integration/Regression Testing, Test Automation
**Tools:** Git, CMake, CI/CD, Oscilloscopes, Logic Analyzers